Completed Fourier path tests: higher-order heat smoothing and integrable control-derivative densities

Project Stardust
A classical formalization companion

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Abstract

Fix a countable family of dyadic path characters and complete their coefficient space using an ℓ^1 quotient. Higher-even-order heat multipliers define normalized continuous expectation functionals and contraction semigroups on this realized Banach space. For every positive smoothing horizon, the orbit under bounded time controls is genuinely Fréchet smooth. Every derivative has a unique Banach-valued Bochner time density, rather than merely a formal distributional kernel. We reconstruct the finite signed-sampler limit, the descent through representation relations, all-order density uniqueness and the scalar realization theorem. An elementary Fourier estimate gives a separately labelled human bound $(n/(epH))^{n/p}$ for the order-n density when the heat order is p. Conjugate-exponent controls have canonical L^p time densities and an explicit uniform exponent boundary. A jointly measurable time/path representative justifies weighted Fubini. A summable cosine example fails to be differentiable without smoothing, and one fixed continuous rough direction excludes a heat expectation from its actual transpose domain. The general Fourier-functional mechanism is classical; comparison with higher-order pseudoprocess and Fresnel literature is explicit. No finite-variation path measure or arbitrary adapted stochastic integral is constructed.

1 The realized domain and the main statement

Throughout, $p = 2m$ with an integer $m \geq 1$. The smoothing parameter H is nonnegative and is not the physical coordinate of a path. Set

$$\mathcal{X} = \{\omega \in C(\mathbb{R}, \mathbb{R}) : \omega(0) = 0\},$$

with the compact-open topology. Choose any countable list of dyadic levels $r_i \geq 0$ and vectors $\xi_i \in \mathbb{R}^{2^{r_i}}$. Adjoin the zero vector as index zero. With $N_i = 2^{r_i}$ and $t_{ij} = (j+1)/N_i$, define

$$\chi_i(\omega) = \exp\left(-2\pi i \sum_{j=0}^{N_i-1} \xi_{ij}\omega(t_{ij})\right). \tag{1}$$

Repeated frequencies and characters are permitted. All characters have modulus one and are continuous. Their coefficient realization is

$$J : \ell^1(\mathbb{N}, \mathbb{C}) \longrightarrow C_b(\mathcal{X}, \mathbb{C}), \qquad Ja = \sum_i a_i\chi_i.$$

The series converges uniformly, and $\|Ja\|_\infty \le \|a\|_1$. Put

$$K = \ker J, \quad \mathcal{A} = \ell^1/K, \quad Q : \ell^1 \to \mathcal{A}, \qquad \|Qa\|_{\mathcal{A}} = \inf_{b:\, Jb = Ja} \|b\|_1 . \tag{2}$$

Thus $\mathcal{A}$ is complete, and its induced realization $\mathcal{J} : \mathcal{A} \to C_b(\mathcal{X})$ is injective and contractive. The quotient norm is part of the theorem: it is not replaced by the uniform norm. An arbitrary selected list need not be closed under frequency addition or negation, so no algebra or conjugation closure is assumed.

Let $U = L^\infty(\mathbb{R}, ds; \mathbb{R})$, and define the control direction

$$h_u(t) = \int_0^t u(s)\, ds \quad (t \in \mathbb{R}).$$

The integral has the usual orientation for negative t. It depends only on the almost-everywhere class of u, is globally Lipschitz, and vanishes at zero. Translation is denoted by τ_h, with realization $\mathcal{J}(\tau_h q)(\omega) = \mathcal{J}q(\omega + h)$.

Define the step functions, extended by zero outside $[0, 1]$,

$$\eta_i(s) = 2\pi \sum_j \xi_{ij} \mathbf{1}_{(0, t_{ij}]}(s), \qquad \lambda_i = \int_0^1 |\eta_i(s)|^p\, ds. \tag{3}$$

Endpoints of the intervals do not affect these classes. The heat evolution will be

$$S_H Qa = Q\big((a_i e^{-H\lambda_i})_i\big). \tag{4}$$

Its well-definedness requires proof; arbitrary coefficient multipliers do not automatically preserve K.

For a real Banach space B, write $B_0 = B$ and $B_{n+1} = L^1(\mathbb{R}; B_n)$, with Bochner integrability. Complex spaces are regarded as real when taking control derivatives. For $u_1, \dots, u_n \in U$, let $\langle u_1 \otimes \cdots \otimes u_n, F\rangle$ denote successive Bochner pairings, starting with the outer variable. Its norm is bounded by $\|F\| \prod_j \|u_j\|_\infty$.

Theorem 1.1. *The formula* (4) *defines a strongly continuous contraction semigroup on $\mathcal{A}$. For $H > 0$ and $q \in \mathcal{A}$, the map*

$$F_q(u) = \tau_{h_u} S_H q \quad (U \to \mathcal{A})$$

is C^∞ over $\mathbb{R}$. For every integer $n \ge 1$, there is a unique $D_{n,H} q \in \mathcal{A}_n$ such that

$$D^n F_q(0)[u_1, \dots, u_n] = \langle u_1 \otimes \cdots \otimes u_n, D_{n,H} q\rangle. \tag{5}$$

These densities are supported in $[0, 1]^n$, in the iterated L^1 sense. The frozen formal construction gives

$$\|D_{n,H} q\| \le (1 + n/H)^n \|q\|_{\mathcal{A}} . \tag{6}$$

The direct human Fourier estimate proved below gives the stronger bound

$$\|D_{n,H} q\| \le C_{n,p,H} \|q\|_{\mathcal{A}} , \qquad C_{n,p,H} = \left(\frac{n}{epH}\right)^{n/p} . \tag{7}$$

For every bounded real-linear $L : \mathcal{A} \to \mathbb{C}$, application of L at every Bochner level gives the unique scalar density of $D^n(LF_q)(0)$, of norm at most $\|L\|\, C_{n,p,H} \|q\|_{\mathcal{A}}$.

The weighted-sampler interpretation is proved in Section 7. This statement concerns every element of the specified completion. It does not identify that completion with all bounded continuous path functions or with a space of predictable integrands. The sharper constant in (7) is not asserted to be a named Lean theorem or the optimal operator norm for every selected spectrum.

2 Actual finite signed samplers and the expectation functional

For $t > 0$, let $k_{p,t}$ be the real Schwartz inverse Fourier transform of $e^{-t(2\pi\xi)^p}$, with Fourier convention $\widehat{f}(\xi) = \int e^{-2\pi i\xi x} f(x)\,dx$. Then

$$\int k_{p,t} = 1, \qquad k_{p,t}(x) = t^{-1/p} k_{p,1}(t^{-1/p}x).$$

Reality follows from the real even multiplier. Schwartz regularity follows because every derivative of the multiplier is a polynomial times a rapidly decreasing function. In particular its inverse transform is integrable. Write $M_p = \int |k_{p,1}|$, a finite number at least one, independent of t after scaling.

At dyadic level r, take $N = 2^r$ signed independent increments with density $\prod_{j<N} k_{p,H/N}(x_j)$, and take the unique Lagrange polynomial path through zero at time zero and the cumulative sums at the dyadic times. This is a continuous path on the whole real line, depending linearly on the increments. This produces a finite signed Borel measure $\nu_{r,H}$ on $\mathcal{X}$, of mass one and total variation M_p^N; the interpolation map is injective, since its increments can be recovered from dyadic evaluations.

The positive sampler $P_{r,H}$ is the pushforward of the normalized absolute increment density. If $\rho(x) = \prod_j k_{p,H/N}(x_j)$, use

$$w_{r,H}(\omega) = M_p^N \operatorname{sgn}_*(\rho(x(\omega))),$$

where $x(\omega)$ recovers the increments, and sgn_* agrees with the sign off zero and has modulus one at zero. The convention at zeros does not affect the sampler. Recovery outside the interpolated-path support may be defined by the same evaluations. Consequently

$$d\nu_{r,H} = w_{r,H}\,dP_{r,H}, \quad |w_{r,H}| = M_p^N, \quad \int w_{r,H}\,dP_{r,H} = 1, \tag{8}$$

and $\int |w_{r,H}| = M_p^N$, $\int w_{r,H}^2 = M_p^{2N}$. This is an actual finite positive sampling law with a signed weight. The positive sampler is generally not a Gaussian proposal law.

Lemma 2.1. *For every i and r, $|\int \chi_i\,d\nu_{r,H}| \le 1$. For $r \ge r_i$,*

$$\int \chi_i\,d\nu_{r,H} = e^{-H\lambda_i}. \tag{9}$$

Proof. At any level, every interpolated evaluation is a real linear combination of the independent increments. Substitution into (1) therefore gives a product of their Fourier transforms, each of the form $e^{-(H/N)(2\pi b)^p} \in (0,1]$. This proves the bound even when some character times are finer than the sampler.

If $r \ge r_i$, all the character times are grid points. In the coarse i-grid, write $\omega(t_{ij}) = \sum_{k\le j} \Delta_k$. Collecting coefficients gives $\sum_j \xi_{ij}\omega(t_{ij}) = \sum_k (\sum_{j\ge k} \xi_{ij})\Delta_k$. Independence and convolution of the finer increments yield

$$\prod_{k<N_i} \exp\left[-\frac{H}{N_i}\left(2\pi \sum_{j\ge k} \xi_{ij}\right)^p\right].$$

This is (9), since η_i equals the corresponding tail sum on each interval of length $1/N_i$. □

Proposition 2.2. *For every $a \in \ell^1$,*

$$\int Ja\,d\nu_{r,H} \longrightarrow \sum_i a_i e^{-H\lambda_i}. \tag{10}$$

The limit depends only on Ja and defines a functional $\mathcal{E}_H : \mathcal{A} \to \mathbb{C}$ of norm exactly one.

Proof. For each fixed r, uniform convergence of the character series allows integration term by term against the finite variation measure $\nu_{r,H}$. Although its variation depends on r, Lemma 2.1 bounds the resulting ith scalar term by $|a_i|$, uniformly in r. Dominated convergence for this counting-measure series proves (10). This uses a character-specific bound, not a nonexistent uniform variation estimate.

If $Ja = 0$, every finite integral on the left vanishes. The limiting scalar series is consequently zero. It annihilates K, so descends to $\mathcal{A}$. Its coefficient bound is $\|a\|_1$; taking the infimum over all representatives gives $|\mathcal{E}_H q| \le \|q\|_{\mathcal{A}}$. The constant character represents $\mathbf{1} \in \mathcal{A}$ with norm one: its coefficient norm is one and its realized supremum norm is one. Its expectation is one, proving the reverse norm inequality. Finite character combinations are dense in the quotient, since finitely supported sequences are dense in ℓ^1. □

3 Translations, heat evolution and the quotient relations

On coefficients define $T_h a = (a_i \chi_i(h))_i$. It is an invertible isometry with inverse T_{-h}, and

$$J(T_h a)(\omega) = Ja(\omega + h).$$

Thus $T_h K = K$, and $\tau_h Q a = Q(T_h a)$ is an isometry of $\mathcal{A}$. Dominated convergence in ℓ^1 proves continuity in h for the compact-open topology: each coordinate $\chi_i(h)$ is continuous, and the coefficient difference is bounded by $2|a_i|$. The same argument, using the finitely many times of each character, proves continuity of $u \mapsto \tau_{h_u} q$ in the L^∞ norm.

Proposition 3.1. *The multiplier in* (4) *preserves K. The induced S_H is a contraction, $S_0 = I$, $S_{H+G} = S_H S_G$, and $S_H \tau_h = \tau_h S_H$. Moreover*

$$\mathcal{J}(S_H q)(\omega) = \mathcal{E}_H(\tau_\omega q), \qquad H > 0. \tag{11}$$

The map $H \mapsto S_H q$ is norm continuous on $[0, \infty)$ for every $q \in \mathcal{A}$.

Proof. For coefficients the right side of (11) is $\sum_i a_i \chi_i(\omega) e^{-H\lambda_i}$, the realization of the proposed evolved coefficients. If $a \in K$, then $T_\omega a \in K$, and Proposition 2.2 makes this scalar zero for every ω. Hence the evolved coefficients belong to K. At $H = 0$ preservation is immediate.

The coefficient multiplier has norm at most one. Its quotient operator is therefore contractive. The remaining algebraic identities follow from multiplication of the diagonal factors, including $e^{-(H+G)\lambda_i} = e^{-H\lambda_i} e^{-G\lambda_i}$, and descend because K is preserved. Finally, if $H_j \to H \ge 0$, the coefficient norm difference tends to zero by domination by $2|a_i|$ and coordinate continuity. Its quotient norm is no larger. This proves strong continuity, without requiring uniform continuity in operator norm. □

There is no positivity assertion in this proposition. The existence of a normalized contraction functional in a coefficient quotient is weaker than existence of a probability path law. Likewise the boundedness of $\mathcal{J}$ does not give a bounded inverse on its range equipped with the uniform norm.

4 Bounded controls separate genuine Bochner time fields

For a real Banach space B, the map

$$P_u : L^1(\mathbb{R}; B) \to B, \qquad P_u f = \int u(s) f(s)\, ds$$

has norm at most $\|u\|_\infty$. Multiplication by u preserves strong measurability and has integrable norm, so this is an ordinary Bochner integral. Bounded linear maps commute with it. Iterating gives the pairing and norm bound used in Theorem 1.1.

Lemma 4.1. *If $F \in B_n$ and all bounded-control pairings of F vanish, then $F = 0$ in B_n. In particular a density satisfying* (5)*, if it exists, is unique as an iterated L^1 class.*

Proof. First let $n = 1$. Taking $u = \mathbf{1}_E$ for every measurable set E of finite measure shows $\int_E F = 0$. A Bochner integrable function is essentially valued in a separable closed subspace $B' \subset B$. There is a countable family (ℓ_j) of norm-one bounded real-linear functionals on B that norms B': choose a countable dense subset of its unit sphere, choose a supporting functional for each point by Hahn–Banach, and use density to obtain $\|x\| = \sup_j |\ell_j(x)|$ for $x \in B'$. The scalar functions $\ell_j(F)$ have integral zero on every finite-measure set, so each vanishes almost everywhere. For example, apply the integral identity on bounded subsets of $\{\ell_j(F) > 1/k\}$ and of $\{\ell_j(F) < -1/k\}$, then let their bounds and k increase. The countable norming family now yields $F = 0$ almost everywhere.

For the induction step, fix the first control u_1. Pairing the outer variable gives an element of B_{n-1} whose every remaining control pairing is zero. It vanishes by induction. Thus all first-control pairings vanish as elements of B_{n-1}. The case $n = 1$, now with Banach target B_{n-1}, gives $F = 0$. Apply this to a difference to obtain uniqueness. □

This proof avoids an uncountable intersection of exceptional sets indexed by all functionals. No reflexivity or Radon–Nikodym property of B is needed. When desired, the iterated spaces identify isometrically with Bochner L^1 on $\mathbb{R}^n$ by the Bochner Fubini theorem. The iterated formulation already defines every contraction and is the one used in the source construction.

5 Native Fréchet derivatives and direct density bounds

Fubini for the finite sum in (3) gives the exact control phase

$$\chi_i(h_u) = e^{-ib_i(u)}, \qquad b_i(u) = \int \eta_i(s)u(s)\,ds. \tag{12}$$

The sign and the factor 2π come from (1); neither can be discarded. Since the support has length one, Hölder's inequality gives

$$\|\eta_i\|_1 \le \|\eta_i\|_p = \lambda_i^{1/p}, \qquad |b_i(u)| \le \lambda_i^{1/p}\,\|u\|_\infty\,. \tag{13}$$

For $n \ge 1$ and $H > 0$, differentiation of a real scalar exponential suggests the coefficient tensor

$$\Big(a_i e^{-H\lambda_i} e^{-ib_i(u)}(-i)^n \prod_{j=1}^{n} b_i(v_j)\Big)_i. \tag{14}$$

We prove both summability and genuine Fréchet differentiability before passing to the quotient.

Lemma 5.1. *For $n \ge 1$,*

$$\sup_{x\ge 0} x^{n/p} e^{-Hx} = \left(\frac{n}{epH}\right)^{n/p} = C_{n,p,H}.$$

The tensor (14) *has ℓ^1 norm at most $C_{n,p,H}\,\|a\|_1 \prod_j \|v_j\|_\infty$. For $n = 0$, use $C_{0,p,H} = 1$.*

Proof. For $x > 0$, the logarithmic derivative of $x^{n/p}e^{-Hx}$ is $n/(px) - H$. The function increases up to $x = n/(pH)$, then decreases to zero, and vanishes at $x = 0$. Substitution yields the maximum. Apply (13) to every factor in (14), sum $|a_i|$, and use this scalar bound. □

Proposition 5.2. *The coefficient orbit $u \mapsto (a_i e^{-H\lambda_i} e^{-ib_i(u)})_i$ is C^∞ from U to ℓ^1. Its nth derivative is (14). The quotient orbit F_q is also C^∞.*

Proof. For real x, y, the integral Taylor formula gives

$$|e^{-i(x+y)} - e^{-ix} + iye^{-ix}| \le |y|^2/2, \qquad |e^{-i(x+y)} - e^{-ix}| \le |y|.$$

At derivative order n, multiply the first inequality by $|a_i|e^{-H\lambda_i} \prod_{j\le n} |b_i(v_j)|$, substitute $y = b_i(w)$, and sum. The remainder in the space of bounded n-linear maps is at most

$$\tfrac{1}{2} C_{n+2,p,H} \|a\|_1 \|w\|_\infty^2 \prod_{j\le n} \|v_j\|_\infty .$$

The candidate derivative is the $(n+1)$st tensor. The second inequality similarly bounds the difference of the nth tensors at $u + w$ and u by $C_{n+1,p,H} \|a\|_1 \|w\|_\infty \prod_j \|v_j\|_\infty$. This proves differentiability and continuity in operator norm, inductively at every order, including order zero. Composition with the bounded quotient map proves the last assertion. In particular the tensors annihilate K after applying Q: if $a \in K$, the entire quotient orbit is identically zero by Proposition 3.1 and translation invariance of K. □

For a representative a, define a field in the iterated ℓ^1-valued L^1 space by the absolutely convergent sum

$$\widetilde{D}_{n,H} a(s_1, \dots, s_n) = \sum_i a_i e^{-H\lambda_i} (-i)^n \eta_i(s_1) \cdots \eta_i(s_n) e_i. \tag{15}$$

Each summand is strongly measurable, supported in $[0,1]^n$, and has integral norm $|a_i|e^{-H\lambda_i} \|\eta_i\|_1^n$. Lemma 5.1 bounds the sum of these norms by $C_{n,p,H} \|a\|_1$. Thus the sum converges in the complete Bochner space; no pointwise convergence on a common exceptional set is required. Its contraction against controls is exactly the coefficient tensor at $u = 0$, by continuity of the pairings.

Proof of the density assertions in Theorem 1.1. Apply Q at each Bochner level to (15); call the resulting field $\mathcal{D}a$. If $a \in K$, every control pairing of $\mathcal{D}a$ is zero by Proposition 5.2. Lemma 4.1 implies $\mathcal{D}a = 0$. Hence $\mathcal{D}$ descends to a bounded linear map $D_{n,H} : \mathcal{A} \to \mathcal{A}_n$. Taking the infimum of the coefficient bound over representatives proves (7). Its control pairing is the derivative of the quotient orbit, establishing (5); uniqueness follows from the same separation lemma. Support is preserved by summation, quotient maps and norm limits.

For a bounded real-linear $L : \mathcal{A} \to \mathbb{C}$, apply its Bochner lift at every depth. Each lift has norm at most $\|L\|$, rather than a new factor at each depth: induction lifts the same previously constructed map, so its norm never increases. Pairing commutes with every lift, yielding L of the tensor in (5). The bounded linear chain rule identifies this with $D^n(LF_q)(0)$. The norm bound and scalar uniqueness follow. □

At a general control u, replace q by $\tau_{h_u} q$ in the zero-control density; its norm is unchanged. Thus these are native Banach-space derivatives at every control. Their multilinear symmetry follows directly from the product in (14), and the density is invariant under permutation of the time variables in its Bochner product-space realization. These are consequences of the explicit formula and uniqueness, not assumptions about formal functional differentiation.

6 The source construction and the total heat horizon

The formal source takes a compositional route. Its first time-gradient map at horizon $T > 0$ is a bounded map

$$G_T : \mathcal{A} \to L^1(\mathbb{R}; \mathcal{A}), \qquad \|G_T\| \le 1 + T^{-1}.$$

It is constructed from the step-function character atoms and descended through the same kernel using bounded-control separation. Its contraction is the genuine first derivative of $u \mapsto \tau_{h_u} S_T q$ at zero. On a generator the representative is $-i\eta_i(s)e^{-T\lambda_i}Qe_i$, agreeing with (15) at $n = 1$.

Iterate G_T using its Bochner lifts. The order-n map has norm at most $(1 + T^{-1})^n$ and contracts to the ordered composition of n smoothed directional derivatives. Each step contributes a new heat factor. On finite generators their product is $e^{-nT\lambda_i}(-i)^n \prod_j b_i(u_j)$. Both the maps and contractions are bounded, so finite-generator density extends this identity to every $q \in \mathcal{A}$. Proposition 5.2, or the source's heat-splitting induction, identifies it with the native nth derivative at total horizon nT.

To obtain the derivative at a prescribed H, one must therefore take $T = H/n$. This yields (6). Taking all n factors at horizon H would instead describe total smoothing nH; it would not prove the stated fixed-horizon identity. The normalized formal density and the direct field of Section 5 have identical contractions, so Lemma 4.1 identifies them. Consequently the human estimate (7) bounds that same canonical density; no new domain or alternative completed space is being substituted.

For $n = 2$, the source also allows two unequal positive horizons S, T. The nested Hessian field has bound $(1 + S^{-1})(1 + T^{-1}) \|q\|$ and contracts to the Hessian at total horizon $S + T$. Pairing it against u, v is a genuine iterated Bochner integral. In the present notation its generator formula is

$$-e^{-(S+T)\lambda_i}\eta_i(s)\eta_i(t)Qe_i.$$

This checks the two minus-imaginary phase factors and the sum, rather than product, of the heat horizons.

7 Realized derivatives and weighted expectation limits

Path evaluation $L_\omega q = \mathcal{J}q(\omega)$ has norm at most one. Its scalar lifted field realizes the nth derivative of

$$u \longmapsto \mathcal{J}(S_H q)(\omega + h_u).$$

This follows from the translation identity and the bounded linear chain rule. It is a derivative of an actual bounded continuous path function in the specified Banach domain.

An expectation functional may have a different even heat order $2k$ and a different positive horizon G. Denote it by $\mathcal{E}_G^{(2k)}$ and use the samplers $P_{r,G}^{(2k)}, w_{r,G}^{(2k)}$. The domain $\mathcal{A}$ depends on the chosen characters, not on the heat order, so this comparison uses the same space. The smoothing densities above still use order p and horizon H.

Theorem 7.1. *Fix $H, G > 0$, $m, k \ge 1$, $q \in \mathcal{A}$, $n \ge 1$ and bounded controls $u_1, \dots, u_n$. At every dyadic sampling level, the weighted realized native derivative is integrable. Moreover*

$$\begin{aligned}\lim_{r\to\infty} \int w_{r,G}^{(2k)}(\omega) D^n\big[v \mapsto \mathcal{J}(S_H q)(\omega + h_v)\big](0)[u_1, \dots, u_n], dP_{r,G}^{(2k)}(\omega) \\ = D^n\big[v \mapsto \mathcal{E}_G^{(2k)}(\tau_{h_v} S_H q)\big](0)[u_1, \dots, u_n] \\ = \langle u_1 \otimes \cdots \otimes u_n, (\mathcal{E}_G^{(2k)})_n D_{n,H} q\rangle.\end{aligned}$$

Here $(\mathcal{E}_G^{(2k)})_n$ is the iterated Bochner lift, not an integral over an unspecified infinite-dimensional signed law.

Proof. Let $a_* = D^n F_q(0)[u_1, \dots, u_n] \in \mathcal{A}$. The realized tensor is exactly $\mathcal{J}a_*(\omega)$ by the bounded evaluation chain rule. Its absolute value is at most $\|a_*\|_{\mathcal{A}}$. Equation (8) now bounds the absolute weighted integrand by $M_{2k}^{2^r} \|a_*\|_{\mathcal{A}}$, a finite constant at every fixed r. This proves integrability.

Apply Proposition 2.2, with order $2k$ and horizon G, to the single completed test a_*. Its finite weighted expectations converge to $\mathcal{E}_G^{(2k)} a_*$. The bounded linear chain rule gives the first identity on the right; commutation with the lifted density pairing gives the second. No derivative is interchanged with a limit using a uniform bound on the weights. Instead the derivative is first established in the completed test space, and the existing continuous expectation-limit theorem is then applied to that test. □

The theorem is pointwise in the chosen controls and derivative order. Its scalar limiting multilinear map has the bound from Theorem 1.1, since $\left\|\mathcal{E}_G^{(2k)}\right\| = 1$. It makes no claim that the weighted densities themselves converge in L^1 under one common probability measure as r varies. The finite sampler laws change with r.

8 Conjugate-exponent controls and a sharp uniform boundary

There is a natural larger local class of controls. Put $q_p = p/(p-1)$ and $V = L^{q_p}(\mathbb{R}; \mathbb{R})$. This notation distinguishes the control exponent q_p from the test element $q \in \mathcal{A}$. Every $u \in V$ is locally integrable and defines the same primitive $h_u(t) = \int_0^t u$. Hölder gives the explicit estimate

$$|h_u(t) - h_u(s)| \le \|u\|_{q_p} |t - s|^{1/p}, \qquad s, t \in \mathbb{R}.$$

Thus h_u is a continuous path. Local absolute continuity also gives $h_u' = u$ almost everywhere, so this direction map is injective on the whole real-line control space. These paths need not be Lipschitz. For example $u(t) = t^{-a}\mathbf{1}_{(0,1]}(t)$ belongs to V whenever $0 < a < 1/q_p$, but is not essentially bounded; its primitive on $[0, 1]$ is $t^{1-a}/(1-a)$ and is not Lipschitz at zero.

For a Banach space B, now define $B^{[0]} = B$ and $B^{[n+1]} = L^p(\mathbb{R}; B^{[n]})$. These are different spaces from the iterated L^1 towers of the main theorem. Hölder defines bounded pairings by controls in V at every level, with the same product norm bound. They separate the towers: all finite-measure indicator controls belong to V, and the local Bochner separation proof of Lemma 4.1 applies. At its first step an L^p field is integrable on every finite-measure set by Hölder; global L^1 integrability is not required for that argument.

Proposition 8.1. *For every $H > 0$ and $q \in \mathcal{A}$, the orbit $u \mapsto \tau_{h_u} S_H q$ is C^∞ on V. Its order-n derivative has a unique time density $D_{n,H}^{[p]} q \in \mathcal{A}^{[n]}$, with*

$$\left\| D_{n,H}^{[p]} q \right\| \le C_{n,p,H} \|q\|_{\mathcal{A}}.$$

The explicit coefficient formula is (15)*, now summed in the iterated L^p space. The native first derivative and its L^p density are formalized in the frozen sources. This all-order L^p tower statement and the sharper constant are human deductions.*

Proof. The phase (12) remains valid for locally integrable controls. Its key bound is now the exact Hölder inequality

$$|b_i(u)| \le \|\eta_i\|_p \|u\|_{q_p} = \lambda_i^{1/p} \|u\|_{q_p}.$$

This substitutes directly for (13) in the Taylor proof of Proposition 5.2. The remainder bounds at orders n and $n+1$ are identical with the V norm in place of the L^∞ norm. They prove actual all-order Fréchet differentiability in this Banach topology, not merely directional differentiability.

The iterated L^p norm of the ith simple tensor in (15) is exactly $|a_i|e^{-H\lambda_i}\,\|\eta_i\|_p^n$. The sum of these norms is at most $C_{n,p,H}\,\|a\|_1$. Completeness therefore gives a genuine Bochner field. Contracting it gives the native tensor; for a coefficient relation in K the tensor is zero in $\mathcal{A}$ because the quotient orbit is identically zero. Separation of these L^p towers annihilates the relation at the field level. Hence the field descends, and taking the infimum over representatives proves the bound. The same separation proves uniqueness. Scalar realization and application of every bounded linear functional follow by Bochner lifting as before. □

The source's first-gradient norm estimate is $1 + H^{-1}$. Its actual closed directional graph, native Fréchet identity and uniquely representing L^p field occur in `FourierPathDualFrechet` and `FourierPathDualTimeGradient`. Proposition 8.1 identifies the latter field with the direct formula by uniqueness. It does not attribute an all-order L^p tower theorem to those first-order declarations.

The bounded-control and conjugate-exponent densities are compatible. They vanish off $[0,1]^n$, and there the continuous embedding of iterated L^p into iterated L^1 has norm at most one, by Hölder on a set of measure one at every level. Truncating any L^∞ control to $[0,1]$ puts it in V without changing any character phase. The embedded field and $D_{n,H}q$ have the same bounded-control pairings; uniqueness in the L^1 tower makes them equal. This uses compact time support, not an invalid global inclusion between $L^\infty(\mathbb{R})$ and $L^{q_p}(\mathbb{R})$.

Proposition 8.2. *The exponent q_p is a sharp lower boundary for a uniform first-gradient bound over arbitrary allowed spectra. More precisely, there is a single admissible countable spectrum such that, for every $1 \le r < q_p$ and every fixed $H > 0$, no finite constant B can bound*

$$\|D[\,u \mapsto \tau_{h_u} S_H q\,](0)[u]\| \le B\,\|q\|_{\mathcal{A}}\,\|u\|_r$$

even just for its single-character tests and bounded step controls.

Proof. For $N = 2^j$, include the character with its only nonzero observation coefficient at $1/N$, chosen so that its tail field is $\eta_N = N^{1/p}\mathbf{1}_{(0,1/N]}$. This is an allowed dyadic character: take the coefficient $N^{1/p}/(2\pi)$ at that first node and zero at all the other nodes. It has $\lambda_N = 1$. Its generator $q_N = Qe_N$ has quotient norm exactly one, by the same realized-supremum and one-coefficient argument used below for terminal characters. Take $u_N = N^{1/r}\mathbf{1}_{(0,1/N]}$, which has L^r norm one and is a bounded step control. The generator derivative equals

$$-ie^{-H} b_N(u_N) q_N, \qquad b_N(u_N) = N^{1/p+1/r-1}.$$

Its norm tends to infinity because $r < q_p$ makes the exponent positive. This contradicts any proposed B. □

Here the single spectrum is fixed and contains all the dyadic spikes. The result does not say that every individual spectrum or test fails at smaller control exponents, or that every such orbit lacks a derivative. It rules out the stated bounded uniform theorem and is a human sharpness deduction from the character formula. All controls used in the obstruction are bounded, so their individual derivatives already exist by Theorem 1.1.

9 Joint time/path representatives and justified Fubini

An almost-everywhere time field evaluated separately at each path does not by itself establish joint measurability. The source supplies this missing bridge for the first L^p density. Let $g = D_{1,H}^{[p]} q \in$

$L^p(\mathbb{R};\mathcal{A})$ and choose a strongly measurable representative $\widetilde{g}:\mathbb{R}\to\mathcal{A}$, equal to g almost everywhere. It can be chosen zero off $[0,1]$. Define

$$j_q(s,\omega)=\mathcal{J}(\widetilde{g}(s))(\omega).$$

The quotient $\mathcal{A}$ is separable, since it is a quotient of countable ℓ^1. The path space $\mathcal{X}$ is Polish for the compact-open topology: $C(\mathbb{R},\mathbb{R})$ is complete and separable for a usual uniform-on- compacts metric, and $\omega(0)=0$ is a closed condition.

Lemma 9.1. *The function j_q is jointly measurable for the product Borel structure, continuous in ω at every fixed s, and satisfies*

$$|j_q(s,\omega)|\le\|\widetilde{g}(s)\|_{\mathcal{A}}.$$

For every $\omega\in\mathcal{X}$ and $u\in V$,

$$\int u(s)j_q(s,\omega)\,ds=\mathcal{J}\big(DF_q(0)[u]\big)(\omega). \tag{16}$$

Proof. The evaluation map $(a,\omega)\mapsto\mathcal{J}a(\omega)$ is jointly continuous. Indeed, at (a_0,ω_0) its difference from $\mathcal{J}a_0(\omega_0)$ is bounded by $\|a-a_0\|_{\mathcal{A}}+|\mathcal{J}a_0(\omega)-\mathcal{J}a_0(\omega_0)|$. Composing this continuous map with the measurable map $(s,\omega)\mapsto(\widetilde{g}(s),\omega)$ proves joint measurability. The stated spaces have their Borel structures; a strongly measurable representative is Borel measurable, and their second-countable structures ensure the product Borel identification. The pointwise bound and path continuity follow from contractivity of $\mathcal{J}$ and its values in $C_b(\mathcal{X})$.

For fixed ω, bounded evaluation commutes with the Bochner integral of $u\widetilde{g}$, which is integrable by Hölder. The defining density contraction is $DF_q(0)[u]$, proving (16). The exceptional time set is that of a single $\mathcal{A}$-valued representative. In particular the argument does not take an uncountable intersection of path-specific null sets. □

Proposition 9.2. *Let μ be an s-finite Borel measure on $\mathcal{X}$ and let real $w\in L^1(\mu)$. For every $u\in V$, the function $w(\omega)u(s)j_q(s,\omega)$ is absolutely integrable under $ds\,d\mu(\omega)$, and*

$$\begin{aligned}&\int\!\!\int w(\omega)u(s)j_q(s,\omega)\,d\mu(\omega)\,ds\\&=\int w(\omega)\mathcal{J}(DF_q(0)[u])(\omega)\,d\mu(\omega).\end{aligned}$$

Its product L^1 norm is at most $\|w\|_{L^1(\mu)}\|u\|_{q_p}\|g\|_{L^p(\mathbb{R};\mathcal{A})}$.

Proof. Lemma 9.1 provides joint measurability and the product majorant $|w(\omega)||u(s)|\,\|\widetilde{g}(s)\|_{\mathcal{A}}$. Its integral is at most the asserted bound by Tonelli and Hölder. Fubini therefore applies to the actual integrable function. Exchange the two integrals and apply (16) for each path to obtain the identity. This establishes the necessary product integrability before any exchange. □

Take $\mu=P^{(2k)}_{r,G}$ and $w=w^{(2k)}_{r,G}$. The bound is finite at every level, although its weight factor $M^{2^r}_{2k}$ can diverge. The proposition identifies the double time/path integral with the weighted expectation of the single completed derivative test. Proposition 2.2 then gives the limit

$$\lim_{r\to\infty}\int\!\!\int w^{(2k)}_{r,G}(\omega)u(s)j_q(s,\omega)\,dP^{(2k)}_{r,G}(\omega)\,ds=\int u(s)\,(\mathcal{E}^{(2k)}_G g)(s)\,ds.$$

The field on the right is the Bochner L^p lift of the expectation functional applied to g. Its pairing equals $\mathcal{E}^{(2k)}_G(DF_q(0)[u])$. The source establishes exactly this first- density sampler limit, not convergence of the unpaired joint fields under a common probability measure. Arbitrary random controls $u(s,\omega)$ are not admitted merely because deterministic controls can be paired. Their products and limits require additional domains and estimates.

10 Two concrete domain and normalization checks

10.1 A two-time character and an invisible constant control

Take the times $1/2, 1$ with frequencies $1, -1/2$. Then η is π on $(0, 1/2]$ and $-\pi$ on $(1/2, 1]$, and $\lambda = \pi^p$. For the constant control $u = \mathbf{1}_{(0,1]}$, $b(u) = 0$. This character is unchanged by the linear direction $h_u(t) = t$ on $[0, 1]$, although its heat multiplier is strictly less than one for $H > 0$.

Let $v = \mathbf{1}_{(0,1/2]}$ and $z = \mathbf{1}_{(1/2,1]}$. Then $b(v) = \pi/2$ and $b(z) = -\pi/2$. At zero path the first v-derivative is $-i(\pi/2)e^{-H\pi^p}$ and the mixed (v, z) derivative is $+(\pi^2/4)e^{-H\pi^p}$. Pairing the density (15) gives these same signs and constants. This example shows why the frequency energy uses all tail sums, rather than just the derivative in a constant-control direction or the terminal frequency alone.

10.2 An unsmoothed orbit that is not differentiable

Choose a spectrum containing the terminal frequencies $j, -j$ for every $j \geq 1$, and put

$$f(\omega) = \sum_{j=1}^{\infty} \frac{\cos(2\pi j\omega(1))}{j^2}.$$

This is an element $q \in \mathcal{A}$: use coefficients $1/(2j^2)$ for the two characters. The coefficient sum is finite and the realization is exactly f. Along $u = \mathbf{1}_{(0,1]}$, write $g(x) = f(h_{xu})$. It is an even continuous real function. With $h_N = 1/(4N)$, all terms in $g(0) - g(h_N)$ are nonnegative, and for $N \leq j \leq 2N$ the angle $2\pi j h_N$ lies in $[\pi/2, \pi]$. Therefore

$$g(0) - g(h_N) \geq \sum_{j=N}^{2N} j^{-2} \geq \frac{N+1}{4N^2} > h_N.$$

If g were differentiable at zero, evenness would force $g'(0) = 0$, contradicting these difference quotients. Evaluation at zero path is a bounded functional on $\mathcal{A}$, so the unsmoothed $\mathcal{A}$-valued control orbit is not Fréchet differentiable there either. This is a concrete element of the completed domain, not merely growth of an operator norm on a sequence of generators.

For a single terminal character of integer frequency j, its quotient norm is exactly one, since its realized supremum norm and its one-coefficient bound are both one. Its unsmoothed control derivative in direction u has norm $2\pi|j|$. Thus no uniform bounded first derivative on all these generators can hold at $H = 0$. Positive heat suppresses these frequencies by $e^{-H(2\pi j)^p}$, exactly the mechanism of Lemma 5.1.

11 A continuous rough direction outside expectation-transfer domains

The preceding sharpness result varies bounded controls inside one spectrum. A distinct source example fixes one continuous direction and still defeats heat-damped expectation transfer. The natural directional operator must first be specified.

For any fixed $h \in \mathcal{X}$, the maps $T_t = \tau_{th}$, $t \in \mathbb{R}$, form a strongly continuous isometric group on $\mathcal{A}$. Define

$$\operatorname{Dom} D_h = \{q \in \mathcal{A} : \lim_{t\to 0}(T_t q - q)/t \text{ exists in } \mathcal{A}\}, \qquad D_h q = \lim_{t\to 0}(T_t q - q)/t.$$

Finite character combinations lie in this domain and are dense. The source identifies this native domain with its actual quotient coefficient graph, rather than postulating derivative values on a larger completion.

For completeness, D_h is closed. For $q \in \operatorname{Dom} D_h$, the group law gives $\frac{d}{dt} T_t q = T_t D_h q$ and hence

$$T_t q - q = \int_0^t T_s D_h q \, ds.$$

The derivative follows by applying T_t to the defining difference quotient; its continuity follows from strong continuity on $D_h q$. The displayed identity is then the Banach-space fundamental theorem of calculus. If $q_j \to q$ and $D_h q_j \to v$, isometry permits passage to the limit under this finite-interval Bochner integral: the error is at most $|t| \, \|D_h q_j - v\|$. Thus $T_t q - q = \int_0^t T_s v \, ds$. Dividing by t and using strong continuity at zero gives $q \in \operatorname{Dom} D_h$ and $D_h q = v$. Density and closedness therefore refer to the original native differentiation graph.

Regard $\mathcal{A}$ as real and let $\mathcal{A}'_{\mathbb{R}}$ be its continuous real dual. Its Banach transpose has the domain

$$\operatorname{Dom} D'_h = \{\phi \in \mathcal{A}'_{\mathbb{R}} : \exists \psi \in \mathcal{A}'_{\mathbb{R}}, \ \phi(D_h q) = \psi(q) \ \forall q \in \operatorname{Dom} D_h\}.$$

The value $\psi = D'_h \phi$ is unique by density. Domain membership is equivalent to an input-norm bound on ϕD_h on the native domain: one implication is continuity of ψ; in the other, the bound gives a unique continuous extension from that dense domain. No graph norm is substituted for the input norm in this criterion.

Proposition 11.1. *There are a fixed countable dyadic spectrum and a fixed continuous direction h such that, for quartic heat and every $H > 0$,*

1. *$D_h S_H$ has no bounded extension agreeing with its values on the finite-character core;*
2. *no continuous real-linear $L : \mathcal{A} \to \mathbb{C}$ satisfies $\mathcal{E}_H(D_h q) = Lq$ throughout $\operatorname{Dom} D_h$;*
3. *the continuous real functional $\phi_H = \operatorname{Im} \mathcal{E}_H$ is not in $\operatorname{Dom} D'_h$.*

Proof. Take $h(t) = (\max(t, 0))^{1/8}$. At level $8n$, put $\delta_n = 2^{-8n}$ and include the character

$$\chi_n(\omega) = \exp(-i 2^{2n} \omega(\delta_n)), \qquad n \geq 0.$$

Its frequency vector has just the first nonzero entry $2^{2n}/(2\pi)$. The associated tail function is $\eta_n = 2^{2n} \mathbf{1}_{(0, \delta_n]}$. For $p = 4$,

$$\lambda_n = 2^{8n} \delta_n = 1, \qquad 2^{2n} h(\delta_n) = 2^n.$$

Its generator q_n has quotient norm one, and direct scalar differentiation gives

$$D_h q_n = -i 2^n q_n, \qquad S_H q_n = e^{-H} q_n, \qquad \mathcal{E}_H q_n = e^{-H}.$$

Consequently $\|D_h S_H q_n\| = 2^n e^{-H} \to \infty$, proving the first claim. Also $\mathcal{E}_H(D_h q_n) = -i 2^n e^{-H}$, whose modulus is unbounded on these unit input tests. This rules out L in the second claim. Its imaginary part is $-2^n e^{-H}$, so it cannot be the value of one bounded real functional ψ on the q_n. The transpose-domain criterion proves the third claim. □

Every generator and its heat evolution in this proof lies in the native directional domain. Unboundedness on that core does not assert that every smoothed test lies outside the domain. The formal rough-transfer and rough-adjoint modules establish the actual second and third exclusions,

with precisely the displayed dyadic spectrum. The first bounded-extension exclusion is an immediate human norm deduction from the same exact source formulas.

This direction is continuous, but its derivative on $(0, 1]$ is $\frac{1}{8}t^{-7/8}$, whose $4/3$ power has divergent integral at zero. It therefore falls outside the local conjugate-exponent control condition. The example is compatible with all the positive theorems above. In particular strong continuity of translation in a continuous direction does not imply that every normalized heat expectation belongs to its transpose domain. A bounded functional and a densely defined closed derivative still require their actual transfer estimate.

12 Antecedents, formal correspondence and limits of the claim

Nishioka's higher-order pseudoprocess calculus [1] already uses Fourier-measure test functionals and weak expectation notions. Its Fourier representations and moment conditions in Sections 1–3, and the higher-order calculus of Sections 5–6, are primary antecedents to the general functional mechanism. The present paper does not claim to originate higher-order heat path calculus. Its chosen countable atomic spectrum, quotient norm and bounded-control Bochner densities require their own domain statements; they are not silently identified with that paper's entire Fourier-measure test class.

Chang, Johnson and Skoug [2], Section 2, describe a classical Banach algebra of Wiener-space functionals represented by finite complex measures on $L^2[0, T]$ through Paley–Wiener–Zygmund integrals. The measure's total variation supplies its norm. Here the characters are pointwise continuous on every path, the spectrum is a fixed countable list, and the realization is quotiented by its actual kernel. Closure under pointwise multiplication was not assumed. Their analytic Wiener/Feynman integral is not the higher-order heat expectation used here.

Mazzucchi, Nicola and Trapasso [3], Section 3 and its Parseval formula, give modern primary context for Fresnel functionals on Fourier transforms of finite complex measures. Their quadratic oscillatory multiplier has modulus one; our multiplier is the real damping $e^{-H\lambda_i}$. No Sjöstrand-class extension or inclusion between their full infinite-dimensional domain and $\mathcal{A}$ is deduced. The sources establish that a Fourier-functional construction, despite failure of an ambient path measure, is an established idea. A complete priority determination for the precise quotient and density package remains unresolved.

The following source correspondence fixes the scope of the formal evidence. The normalization, finite-sampler, translation and density constructions are reconstructed above; the semigroup strong-continuity argument in Section 3 is a human dominated-convergence deduction. All files are under `Foundation/` at frozen commit `bab688bb6140cd6316c80680095a914ea0d346db`; exact declarations, line numbers and source hashes are supplied separately in `theorem-map.json`.

- `HigherHeatFourierBanach`: closed-kernel quotient, completeness, bounded injective realization, norm-one expectation and dense finite generators. `HigherHeatMixedFourierPaths`: character bounds, completed series limits and representation independence.

- `HigherHeatWeightedPaths`: actual signed density over the absolute-density probability sampler, its exact first and absolute moments, and finite-level integrability.

  `HigherHeatFourierPathSemigroup`: multiplier kernel preservation and contraction semigroup.

- `FourierPathControlFrechet` and

  `FourierPathSmoothControls`: native real Fréchet differentiability and all-order smoothness on L^∞ controls.

`BochnerTimeControlPairing` and

`BochnerControlTowers`: actual Bochner pairing, norm bounds and separation.

- `FourierPathTimeTowers`,

  `FourierPathTimeTensorIdentity` and

  `FourierPathTimeTensorRealization`: composed densities, total-horizon normalization, uniqueness, scalar realization and the weighted density limit. `FourierPathHessianRealization` gives the unequal-horizon two-time specialization.

The conjugate-exponent all-order tower and its uniform exponent boundary are human deductions; the named formal declarations in that family give the native first derivative and its L^p field. The joint representative and weighted Fubini bridge are formalized in `FourierPathDualJointDensity`, `FourierPathDualDensityFubini` and
`FourierPathDualDensitySamplers`.

The fixed rough direction, native closed directional graph and transpose exclusions are supplied by `FourierPathRoughDirection`,
`FourierPathRoughTransferObstruction`,
`FourierPathRoughAdjointObstruction` and the directional-generator modules. The source also proves strong continuity in `HigherHeatFourierPathC0`; the direct proof in Section 3 is consistent with that declaration.

The direct bound (7), the explicit cosine obstruction, the two-time calibration, and the direct all-order coefficient proof are human reconstructions or deductions. The formal bound being cited is (6); the stronger estimate is not attributed to a declaration that proves only the coarser bound. The source proof review is specific to the mapped modules, not a claim that all imported library proofs were reread.

For $p > 2$, the one-time signed kernel changes sign and $M_p > 1$. Finite-grid variation then grows as $M_p^{2^r}$, obstructing a finite-variation path measure with all these marginals. The separate signed multichannel companion reconstructs that classical obstruction in detail. The growth is compatible with Proposition 2.2, whose bound exploits individual Fourier characters. It does not become a probability law through the positive finite samplers of (8). At $p = 2$ the Gaussian case has $M_2 = 1$, while all control-density assertions above remain valid.

Finally, there is no construction here of a conditional expectation with respect to an infinite signed law, a martingale filtration, an Itô isometry, or an integral of arbitrary adapted random controls. The Banach domain and positive smoothing horizon are essential assumptions. Every claimed derivative is a native derivative of its specified orbit, and every displayed time integral is a Bochner integral of an established integrable density.

References

[1] K. Nishioka, *Stochastic calculus for a class of evolution equations*, Japanese Journal of Mathematics, New Series **11** (1985), 59–102. Primary publisher article and full text.

[2] K. S. Chang, G. W. Johnson and D. L. Skoug, *The Feynman integral of quadratic potentials depending on two time variables*, Pacific Journal of Mathematics **122** (1986), no. 1, 11–33. Primary full text, especially Section 2.

[3] S. Mazzucchi, F. Nicola and S. I. Trapasso, *Phase space analysis of finite and infinite dimensional Fresnel integrals*, arXiv:2403.20082, version 2 (2025). Primary preprint, especially Section 3.